# Week-2 Essay

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## Introduction

In *Weapons of Math Destruction* (WMD), Cathy O'Neil warns of the dangers of data-driven algorithms, particularly when they are used uncritically in the workplace. In Chapter 4, she focuses on how these models impact hiring and employee evaluations, perpetuating biases that reinforce social inequalities. Similarly, in *Everybody Lies*, Seth Stephens-Davidowitz explores how data science can reveal uncomfortable truths about human behavior that challenge our gut instincts. His first two chapters highlight how big data can validate or refute psychological theories and human assumptions. Through their works, O'Neil and Stephens-Davidowitz underscore the importance of transparency and accountability in data science to ensure these models benefit society rather than perpetuate harm.

## Literature Review

O'Neil's Chapter 4, "Propaganda Machine," critiques how algorithms used in hiring often have hidden biases. The chapter begins with the author's experience at an advertising start-up, where a venture capitalist promotes the future of personalized advertising. O'Neil critiques this vision, highlighting how for-profit colleges exploit data to target vulnerable individuals seeking upward mobility, often leaving them with unmanageable debt. The chapter explores how these institutions use data analytics to identify and prey on potential students, promising false hope of prosperity. It argues that such practices perpetuate inequality and calls for greater transparency and regulation to protect consumers (O'Neil, 2016). O'Neil argues that without feedback loops or accountability, these models can unfairly penalize people and restrict social mobility.

In Chapter 1, "Your Faulty Gut," Stephens-Davidowitz argues that intuition, or "gut feelings," often leads people to incorrect conclusions about human behavior. He explains that while we might trust our instincts, they frequently fail when faced with complex patterns that only data can accurately reveal. He contrasts these flawed instincts with the power of big data, which can uncover hidden truths and identify patterns that go against conventional wisdom (Davidowitz, 2017). For example, big data can more accurately reflect trends in health and social behavior, areas where our assumptions may be clouded by biases and incomplete information. The chapter argues that while intuition is valuable, data science provides a more precise understanding of the world, challenging the belief in relying solely on gut instincts.

In Chapter 2, "Was Freud Right?", Stephens-Davidowitz explores how big data can validate—or question—Freud's theories on the subconscious mind and repressed desires. He uses data from Google searches and social media interactions to examine modern attitudes toward taboo subjects, observing that people reveal hidden biases, fears, and desires online that they may never share in person. Stephens-Davidowitz describes this digital activity as a "truth serum," as it often reveals genuine thoughts and behaviors that people would otherwise suppress (Davidowitz, 2017). Through this exploration, he suggests that while Freud's theories may have some merit, big data provides a more nuanced understanding of human psychology, illuminating aspects of our psyche that were previously hidden or misrepresented in traditional studies.

## Analysis of the Issue

O'Neil argues that hiring and evaluation algorithms function as "black boxes," making it difficult for affected individuals to understand or question their workings (O'Neil, 2016). These models often prioritize standardized metrics that overlook a candidate's full capabilities, instead focusing on proxies like credit scores, which can disadvantage individuals from lower socioeconomic backgrounds and create a "downward spiral" that reinforces inequality (O'Neil, 2016). She calls for greater algorithmic accountability to prevent these models from perpetuating social inequities.

In contrast, Stephens-Davidowitz sees value in big data's potential to reveal hidden truths when responsibly analyzed. He explains how data can challenge gut instincts and expose real behaviors and attitudes that people often conceal, especially around sensitive topics like prejudice and relationship anxieties (Davidowitz, 2017). Using big data as a "digital truth serum," he suggests, allows society to gain insights into human psychology, even if these truths are sometimes uncomfortable (Davidowitz, 2017).

## Conclusion and Limitations

Both authors argue for a more ethical and transparent use of big data, although their perspectives on its potential differ. O'Neil calls for increased accountability in algorithms to prevent models from "punishing the poor" and to protect against the reinforcement of social inequalities (O'Neil, 2016). She believes that algorithms must incorporate feedback mechanisms to ensure fairness and prevent harm. Meanwhile, Stephens-Davidowitz sees the value of big data as a tool for gaining insights that go beyond "faulty gut" assumptions and social taboos, but he also notes that big data can be prone to ethical risks if not carefully managed (Davidowitz, 2017). Ultimately, both authors advocate for responsible data practices, emphasizing the need to balance innovation with a commitment to equity and transparency.

## References

Davidowitz, S. S. (2017). *Everybody lies: Big data, new data, and what the internet can tell us about who we really are.* Dey Street Books.

O'Neil, C. (2016). In *Weapons of math destruction: how big data increases inequality and threatens democracy.* Crown Publishers.